ISAM ABDULLAH BALGHARI

Computational Physicist & Software Engineer

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SUMMARY

Computational physicist with six years of production software experience in computer vision and deep learning, specialising in E(3)-equivariant machine learning for quantum materials simulation. Author of DSpinGNN, a physics-informed equivariant GNN for mesoscale spin-lattice dynamics in strain-engineered 2D magnets, submitted to Computational Materials Science (Elsevier)

EDUCATION

MS Physics

Lahore University of Management Sciences

09/2023 - 07/2026

- Supervisor: Dr. Muhammad Sabieh Anwar
- Co-supervisor: Dr. Muhammad Faryad
- Thesis: DSpinGNN: A Physics-Informed Equivariant Graph Neural Network for Dynamic Magnetic Exchange Prediction in Strain-Deformed Monolayer CrI_3 (submitted, Computational Materials Science (Elsevier))
- COURSEWORK: Density Functional Theory — online course, Coursera (certificate) | Quantum Information Science & Technology A+ | Quantum Computing A+ | Quantum Algorithms A+ | Electrodynamics A | Statistical Mechanics A | Electrochemical Energy Storage & Conversion A

BS Computer Science

Lahore University of Management Sciences

09/2013 - 06/2017

- Capstone: Visual Inertial Odometry — robust localization via Multi-State Constraint Extended Kalman Filter (MSCKF), fusing visual feeds with kinematic robot models
- Research: Funded research internship, University of Siegen, Germany — computational experiments for spatial stereo setups
- Relevant Coursework: Machine Learning, Applied Probability, Applied Statistics, Applied Stochastic Processes

PUBLICATIONS

DSpinGNN: A Physics-Informed Equivariant Graph Neural Network for Dynamic Magnetic Exchange Prediction in Strain-Deformed Monolayer CrI_3

Preprint at arXiv: arxiv.org/abs/2606.11685

I. A. Balghari, M. Faryad, and M. S. Anwar

arxiv.org/abs/2606.11685

PhysTrackX: An Open-Source Framework for Automated Tracking and Sensor Synchronization

Apparatus Note in final preparation for submission to the American Journal of Physics (2026)

I. A. Balghari and M. S. Anwar

SKILLS

PyTorch | e3nn | NequIP | Geometric Deep Learning E(3)-Equivariant Graph Neural Networks | e3nn (equivariant convolutions, tensor products, spherical harmonics)

Computational Physics

Density Functional Theory (DFT+U) | Quantum ESPRESSO | Wannier90 | TB2J | SLURM | Statistical Mechanics | ASE | OVITO | FIRE relaxation

Software & Systems

Python | C/C++ | Git | Linux & Systems | Pipeline architecture | Compute optimisation

AWARDS

Dean's Honour Roll
Awarded for outstanding CGPA maintained during MS studies

LUMS Merit Scholarship
Awarded for exceptional academic performance during MS Physics; competitive departmental scholarship

Facebook Hacker Cup - Qualification Round
Qualified for international algorithmic programming competition

Google Code Jam - Qualification Round
Qualified for international algorithmic programming competition.

CUSTOM

Teaching Experience

Computer Vision - Lahore University of Management Sciences - Spring 2017
Delivered tutorial sessions and office hours for junior students on visual feature extraction, epipolar geometry, and matrix operations underlying modern visual algorithms; supported course delivery for one semester during BS enrolment.

Quantum Machine Learning - Lahore University of Management Sciences - Spring 2025
Delivered tutorial sessions and supported instruction of a graduate-level course integrating quantum information theory with modern machine learning; assisted students with problem sets spanning variational quantum circuits, quantum kernel methods, and hybrid classical-quantum architectures

Independent Project: (Software Released)

Scientific Software (with Dr. M. S. Anwar): Designed and shipped PhysTrackX, an open-source video-tracking and sensor-synchronization framework for experimental physics, now used in teaching laboratories (16 releases, AJP apparatus note in preparation). [code]

EXPERIENCE

Graduate Researcher

Lahore University of Management Sciences

2023 - 2026 Pakistan

Sabieh Anwar Research Group, LUMS, Pakistan Thesis project: DSpinGNN — Physics-Informed Equivariant GNN for Dynamic Magnetic Exchange Prediction in Strain-Deformed CrI_3 (submitted, Computational Materials Science (Elsevier))

- Developed DSpinGNN [code], a bifurcated E(3)-equivariant architecture coupling a NequIP-inspired GNN (implemented using e3nn tensor product operations and PyTorch) for Langevin structural dynamics with a physics-informed Δ-MLP exchange predictor that embeds the Goodenough-Kanamori superexchange relationship as a closed-form analytical inductive bias — making predictions interpretable in terms of established quantum chemistry
- Achieved test-set MAEs of 1.1 meV/atom (energy), 6.5 meV/Å (force), and 0.18 meV (exchange coupling, $R^2 = 0.91$) on a stratified 61-configuration held-out set, with all three quantities evaluated simultaneously against DFT+U ground truth
- Demonstrated 400× system size transferability without retraining: deployed an 8-atom-trained model on a 3,200-atom supercell, driving stable Langevin dynamics at 5 K and extracting mesoscale observables inaccessible to direct DFT — domain wall width $\xi = 1.7 \pm 0.3$ nm and interference oscillation period $\tau = 0.27$ ps, consistent with experimentally verifiable mesoscale features
- Built SpinDFT [code], an end-to-end HPC orchestrator automating ~470 DFT+U → Wannier90 → TB2J calculations with dynamic Fermi-level calibration, crash-proof NSCF diagonalization, and just-in-time wavefunction deletion; released as an independent open-source repository alongside DSpinGNN
- Reproduced DTNN and NequIP using PyTorch + e3nn as foundational preparation, independently implementing equivariant message passing, tensor field operations, and force prediction heads without reference to the authors' codebases

Research Intern

University of Siegen

2016 - 2016 Germany

- Executed computational experiments for spatial stereo vision setups as part of a funded international research placement; gained hands-on experience with 3D reconstruction pipelines and multi-camera calibration

Senior Software Engineer

Roll Camera Inc

02/2021 - 02/2025 Remote

- Overhauled the core deep learning inference engine, reducing total compute time by 60× through pipeline architecture restructuring and operator-level optimisation
- Designed and deployed an AI-driven 3D visual effects system end-to-end, establishing deployment standards that accelerated enterprise product shipping by 70%. [Sample]
- Resolved critical architectural bottlenecks under strict production deadlines to stabilise high-demand features for early enterprise adoption

Independent Engineer

01/2020 - 01/2021 Remote

- Independently conceived and deployed an AI-driven spatial tracking pipeline for 3D vehicle detection and kinematic analysis in heavily occluded environments — self-directed research bridging industry CV work and formal academic inquiry. [Code](https://github.com/isamabdullah88/RDTracking) [Sample]

Software Engineer

Vidimensions

01/2019 - 12/2019 Singapore

- Engineered a statistical ensemble fusion methodology combining multiple deep learning architectures for complex spatial analytics tasks
- Improved anomalous behaviour detection accuracy by 10% through enhanced mathematical modelling of spatial decision boundaries

Software Engineer

Hazen.ai

08/2017 - 12/2018 Pakistan

- Designed and implemented non-linear Kalman filters for object kinematics modelling, improving trajectory tracking precision by 20%. [Sample]
- Led architectural development of a 3D vehicle detection prototype deploying spatial constraints for occlusion-robust performance

CERTIFICATIONS

Density Functional Theory
Coursera

Matrix Algebra for Engineers
Coursera

Differential Equations for Engineers
Coursera

Quantum Mechanics for Engineers
Coursera